BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022 • Paper Code: DIK-HSSC-XII-2022 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 20 | **Paper Code:** DIK-HSSC-XII-2022 (A)

Q1. Choose the correct option i.e. A,B,C,D and fill it up in the correct bubble in the given MCQs sheet.

Qi. If the point A, B, C are collinear then area of triangle region is_____

(A) 2 (B) 1

(C) 0 (D) ∞

Qii. The radius of circle $x^2 + y^2 - 8x - 6y + 9 = 0$ is _____

(A) 3 (B) 4

(C) 16 (D) 19

Qiii. $y^2 = 4px$directrix is = _____

(A) $x = -p$ (B) $x = p$

(C) $y = p$ (D) $y = -p$

Qiv. A line $y = mx + c$ is tangent to circle $x^2 + y^2 = a^2$ if c = ___

(A) $\pm\sqrt{(a^2m^2)}$ (B) $\pm\sqrt{(m^2 - b^2)}$

(C) $\pm a\sqrt{(m^2 + 1)}$ (D) $\pm\sqrt{(m^2 - 1)}$

Qv. The order of different equation $d^3y/dx^3 + [d^2y/dx^2] - 8\ dy/dx + 2y = 8$, is_____

(A) 2 (B) 3

(C) 1 (D) 4

Qvi. $f(x,y) = x^3+xy^2$, then $f(0,-1)$ = _____

(A) 1 (B) 2

(C) 0 (D) −1

Qvii. The regular Falsi'method is _____ interactive method

(A) One point
(B) Zero point
(C) Three point
(D) Two point

Qviii. if $\lim(x\to c)(f(x)) = L$, the L is called the _____ function f(x)

(A) LHL
(B) RHL
(C) Limit
(D) None [illegible in source scan]

Qix. $|x| = x$ if_____

(A) $x = 0$
(B) $x < 0$
(C) $x \geq 0$
(D) x = [illegible in source scan]

Qx. $d/dx(x^n)$ = _____

(A) nx^{n-1}
(B) x^{n-1} [illegible in source scan]
(C) nx^n [illegible in source scan]
(D) x^n [illegible in source scan]

Qxi. d/dx(tanx) = _____

(A) Tan x
(B) Sec x
(C) Sec^2x
(D) Secx [illegible in source scan]

Qxii. The midpoint of line segment joining points (a, −b) and (−a, b) is _____

(A) (2a, 0)
(B) (0, 0)
(C) Parabola
(D) Hyperbola [illegible in source scan]

Qxiii. The graph of function f''(x) is concave downward in interval (a, b) if _____

(A) $f''(x) > 0$
(B) $f'(x) > 0$
(C) $f''(x) = 0$
(D) $f''(x) < 0$

Qxiv. $\int(8x-1)/(x^2-x-2)\,dx$, the integrand is _____ fraction

(A) Proper
(B) Improper
(C) Compound
(D) Decomposable [illegible in source scan]

Qxv. The domain of vector function f(t) = 2ti − 3tj [illegible in source scan] is_____

(A) R
(B) W [illegible in source scan]
(C) R−{0}
(D) Z [illegible in source scan]

Qxvi. $\int_1^4 3x^2\,dx$ = _____

(A) 63
(B) 21
(C) 4
(D) 18 [illegible in source scan]

Qxvii. $\int dx/(x+1) =$ _____

(A) ln(x + 1)
(B) ln x + c
(C) ln x
(D) ln(x−1) [illegible in source scan]

Qxviii. The point p(10, −6) is _____ line 9x + 10y − 3 = 0

(A) Above
(B) Below
(C) Lying
(D) None

Qxix. Slope of the line ax + by + c = 0 is _____

(A) a/b
(B) −a/b
(C) b/a
(D) −b/a

Qxx. Two lines represented by $ax^2 + 2hxy + by^2$ are parallel if

(A) ab = h
(B) $ab + h^2 = 0$
(C) ab − h = 0
(D) $h^2 - ab = 0$

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 50

Q2. Attempt any TEN parts. Each part carries equal marks.

i. Discuss the continuity of $f(x) = (x^2-4)/(x+2)$ at x = −2

ii. Evaluate limit of sequence: {(5n)/(n+7)}

iii. Find the derivative of $f(x) = x^2 + 1$, using first principle rule.

iv. y = e^√(x−5), find dy/dx

v. $x = 4t^2 + 1$, $y = 6t^3 + 1$, find d^2y/dx^2

vi. Evaluate: $\int dx/(x^2-a^2)$

vii. $\int_1^2 (5t^2-3t+18)/(t(9-t^2))\, dt$

viii. Find out the area between the curve f(x) and x-axis, such that $f(x) = 4 - x^2$, [0, 3].

ix. Find the equation of the straight line passing through points A(1, 0) and B(2, 5).

x. For what value of 'n' the line x + y + n = 0 touch the circle $x^2 + y^2 = 9$.

xi. Find the equation of tangent and normal to parabola $y^2 = 12x$ at point (3, 6).

xii. Solve: dy/dx = cos x, using initial condition, y(0) = 1

xiii. $f(x,y) = \sin(x^2) \cos y$, find f_x and f_y

SECTION – C

Maximum Marks: 30

Note: Note: Attempt any Three Questions. All questions carry equal marks:

Q.3 a. Find all the relative extrema of $f(x) = x^3 + 6x^2 + 9x + 2$

b. Determine the equation of the circle having center at point $C(-9, -6)$ and circumference passes through origin.

Q.4 a. Find the area of triangle region whose vertices are $A(4, -5)$, $B(5, -6)$ and $C(3, 1)$.

b. Find maclaurin series expansion for $f(x) = \ln(1 - 4x)$

Q.5 a. Find the equation of ellipse with vertices$(-5, 0),(5, 0)$ and eccentricity $e = 3/5$

b. Evaluate by parts $fe^x \cos x dx$

Q.6 a. If $z = zy\, f[y/x]$, then show that $x\, \partial z/\partial x + y\, \partial z/\partial y = 2z$

b. Use Trapezoidal rule to approximate the value of $\int_1^3 x^2\, dx$, $n = 4$